

August 17, 2026

1 Hierarchical TD3 EV-GNN for Large-Scale Public Power Set-point Tracking

1.1 Four-Scale Analysis of Infrastructure-Aligned Decision Formation

1.1.1 Research question

Under the same TD3 framework, EV2Gym PublicPST task, graph-structured state representation, reward family, and charging-only EV action domain, does explicitly factorising charging decisions through the CPO-Transformer-Charger-EV infrastructure hierarchy improve control performance and operational behaviour compared with direct EV-node action formation?

1.1.2 Models compared

- **TD3 EV-GNN Baseline with PublicPST Charging-Only Action Mapping** — shortened in figures to **TD3 EV-GNN Baseline**.
- **Hierarchical TD3 EV-GNN with CPO-Transformer-Charger-EV Decision Allocation** — shortened in figures to **Hierarchical TD3 EV-GNN**.

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The formal experiment contains **4 PublicPST scenarios** $\times$ **2 architectures** $\times$ **10 independently trained paired seeds** = **80 trained models**. Each model was trained for **75,000 environment steps**, with scheduled deterministic evaluation every **5,000 steps**. Training behaviour is examined before the selected `model.best` policies are compared in the final-performance analysis.

Primary domain reference: Orfanoudakis et al. (2025), *Scalable reinforcement learning for large-scale coordination of electric vehicles using graph neural networks*, *Communications Engineering*, 4, 118. <https://doi.org/10.1038/s44172-025-00457-8>

TD3 reference: Fujimoto, van Hoof & Meger (2018), *Addressing Function Approximation Error in Actor-Critic Methods*. <https://arxiv.org/abs/1802.09477>

2 PublicPST operating problem and the 1000-charging-point case study

Public Power Setpoint Tracking (PublicPST) is easiest to understand as an **aggregate control problem**. The Charging Point Operator (CPO) does not simply ask whether one EV is charging. At each 15-minute control step, it must coordinate many connected EVs so that the **total charging power of the fleet** follows a requested power trajectory.

At control step t ,

$$P_t^{\text{set}}$$

is the aggregate charging power requested by the PublicPST task, while

$$P_t^{\text{realised}}$$

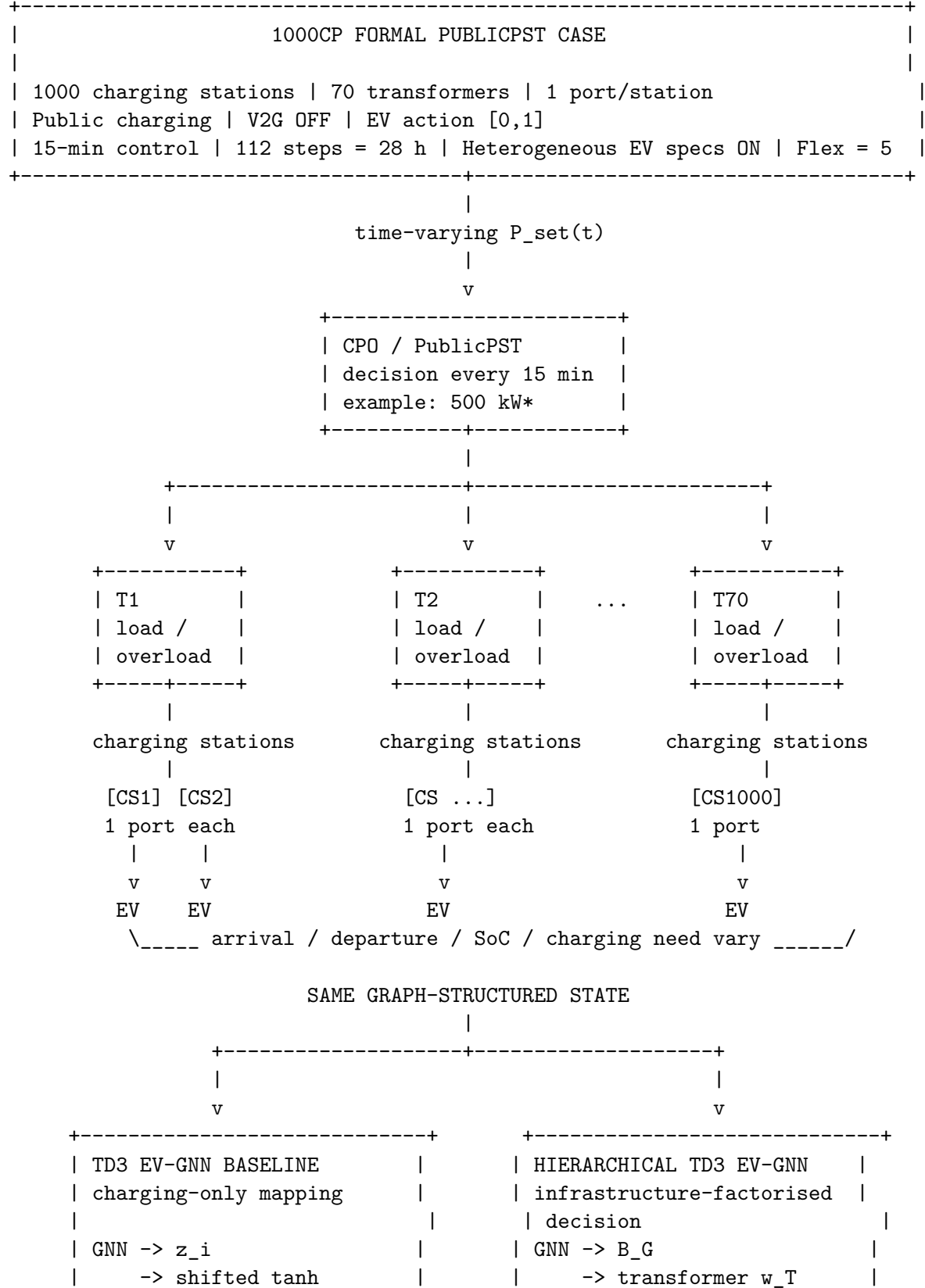
is the aggregate charging power produced after the controller’s EV actions are applied. A signed tracking mismatch can therefore be written as

$$e_t = P_t^{\text{set}} - P_t^{\text{realised}}.$$

For one **illustrative control interval only**, suppose $P_t^{\text{set}} = 500$ kW and the fleet realises $P_t^{\text{realised}} = 470$ kW. The fleet is then 30 kW below the requested aggregate level at that instant. The 500 kW value is used only to make the control story concrete; the formal PublicPST setpoint changes through the simulated episode.

The same decision matters to three different stakeholders. For the **CPO**, closer tracking means that the aggregate charging fleet follows the requested power trajectory more reliably. For the **distribution infrastructure**, good fleet-level tracking is incomplete if the controller creates excessive loading on particular transformer branches. For **EV users**, infrastructure protection is also incomplete if it is achieved by simply withholding charging service. The analysis therefore treats **control efficacy, infrastructure consequences, and EV-service outcomes as related but distinct questions**.

2.1 Figure 1 - 1000CP PublicPST case study and policy decision architecture



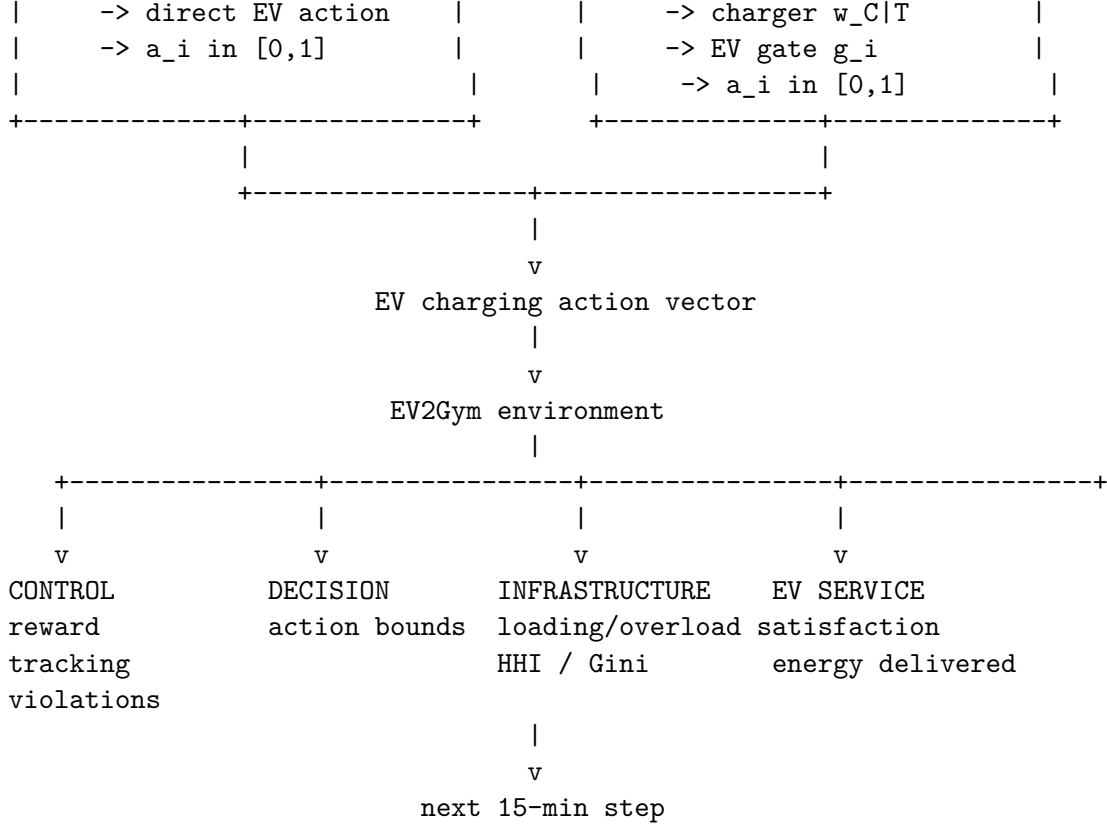


Figure 1. Read the diagram from top to bottom. The upper half describes the **actual 1000CP formal scenario**: 1000 single-port charging stations are connected through 70 transformer branches, weekday episodes use random-day sampling, EV specifications are heterogeneous, EV availability changes with arrivals and departures, and the CPO receives a time-varying aggregate PublicPST request every 15 minutes. The value **500 kW*** is an illustrative single-interval request used only to explain the control loop; it is not a fixed setpoint in the experiment.

The lower half isolates the research intervention. Both policies observe the **same graph-structured network state** and operate through the same charging-only $[0, 1]$ EV action interface. The TD3 EV-GNN baseline uses the graph to form EV actions directly. The hierarchical actor uses the same graph information but makes the physical hierarchy part of the decision itself: it first allocates across transformers, then across chargers within each transformer, and finally applies an EV-specific gate.

In plain language, imagine one 15-minute interval in this 1000-station network. Some EVs have just arrived, some are close to departure, and their states of charge and remaining charging needs differ. The CPO must turn that changing network state into many simultaneous charging commands. The experiment asks whether it is enough for the actor to **know** the infrastructure topology through the GNN, or whether the actor should also **use that topology explicitly when forming the final decision**. Once the actions are applied, the same EV2Gym environment produces the outcomes analysed later: tracking performance, action-boundary behaviour, transformer stress, allocation concentration, and EV-service outcomes.

The 1000CP configuration also contains a transformer **max-power** setting of **100** in the formal YAML. It is treated here only as a simulator configuration value; no engineering unit or physical nameplate rating is inferred beyond what the configuration explicitly defines.

2.1.1 Formal action formation

Figure 1 shows the structural difference; the equations below state the same difference precisely.

For the TD3 EV-GNN baseline, each EV-node actor logit z_i is mapped into the charging-only PublicPST action range using

$$a_i^{\text{base}} = \frac{a_{\text{max}}}{2} [\tanh(z_i) + 1], \quad a_i^{\text{base}} \in [0, a_{\text{max}}].$$

The hierarchical actor retains the same graph encoder but factorises the final EV decision. For graph G ,

$$B_G = N_{\text{active EV}},$$

where $N_{\text{active EV}}$ is the number of active EV nodes in that graph. Importantly, B_G is a **normalised actor scale**, not a physical kW dispatch budget.

Transformer allocation is formed with a graph-wise softmax,

$$w_{T(j)} = \frac{\exp(s_{T(j)})}{\sum_{k \in \mathcal{T}_G} \exp(s_{T(k)})},$$

so the transformer weights express how decision authority is divided across the active infrastructure branches. Charger allocation is then normalised **within the parent transformer**,

$$w_{C(i)|T(i)} = \frac{\exp(s_{C(i)})}{\sum_{c \in \mathcal{C}_{T(i)}} \exp(s_c)}.$$

Finally, the EV-specific gate is

$$g_i = \sigma(s_i),$$

and the EV action is

$$a_i^{\text{hier}} = \text{clip} \left(B_G w_{T(i)} w_{C(i)|T(i)} g_i, 0, a_{\text{max}} \right).$$

Non-EV graph rows remain zero in the action tensor. The central distinction is therefore **not whether topology is available to the policy**—both models encode the graph. The distinction is whether transformer and charger structure is also used to organise the final action.

2.1.2 Minimal code-level comparison

```
# TD3 EV-GNN baseline: graph encoding -> direct EV action
node_embeddings = gnn(graph_state)
raw_logits = action_head(node_embeddings)
ev_action = 0.5 * max_action * (torch.tanh(raw_logits) + 1.0)
```

```

# Hierarchical TD3 EV-GNN: graph encoding -> infrastructure-factorised EV action
node_embeddings = gnn(graph_state)
transformer_weight = graph_softmax(transformer_scores)
charger_weight = grouped_softmax(charger_scores, parent_transformer)
ev_gate = torch.sigmoid(ev_gate_logits)

ev_action = (
    active_ev_count
    * transformer_weight
    * charger_weight
    * ev_gate
).clamp(0.0, max_action)

```

These fragments summarise the decision-forming logic in TD3/TD3_ActionGNN_NonNegative.py and TD3/TD3_HierarchicalActionGNN.py. They are included to make the architectural contrast explicit, not as replacement implementations.

```

[1]: # Numerical illustration of action formation only; these values are not
    ↪ experimental results.
import numpy as np

max_action = 1.0

baseline_logit = 0.40
baseline_action = 0.5 * max_action * (np.tanh(baseline_logit) + 1.0)

active_evs = 4
transformer_weight = 0.30
charger_weight = 0.50
ev_gate = 0.60
hierarchical_action = np.clip(
    active_evs * transformer_weight * charger_weight * ev_gate,
    0.0,
    max_action,
)

print(f"Illustrative baseline action:      {baseline_action:.3f}")
print(f"Illustrative hierarchical action: {hierarchical_action:.3f}")

```

```

Illustrative baseline action:      0.690
Illustrative hierarchical action: 0.360

```

The numerical illustration makes the two pathways concrete for one EV. With the illustrative baseline logit used above, the baseline forms the EV action directly through the shifted-tanh mapping. For the hierarchical actor,

$$a_i = \text{clip}(4 \times 0.30 \times 0.50 \times 0.60, 0, 1) = 0.36.$$

The two numerical actions are **not performance evidence** because they were generated from

different illustrative intermediate values. Their only purpose is to show that the baseline exposes a direct EV-level output, whereas the hierarchical actor exposes intermediate transformer, charger, and EV-gating decisions before the final command is formed.

2.1.3 What changed and what was held fixed

Component	TD3 EV-GNN Baseline	Hierarchical TD3 EV-GNN	Role in comparison
EV2Gym PublicPST environment	Same	Same	Controlled
TD3 optimisation framework	Same	Same	Controlled
Graph-structured state	Same	Same	Controlled
Reward family	Same	Same	Controlled
Critic/replay family	Same	Same	Controlled
EV charging action domain	[0, 1]	[0, 1]	Controlled
Final decision formation	Direct EV-node action	Transformer allocation -> charger allocation -> EV gate	Research variable

This design isolates **decision factorisation**. It does not test whether one reinforcement-learning algorithm is intrinsically better than another; TD3, the PublicPST environment, graph state, reward family, critic/replay family, and final charging-only action domain are deliberately held fixed.

3 Why the 75,000-step aligned experiment was required

The earlier 50,000-step experiment was useful because it generated two substantive hypotheses: explicit infrastructure-aware decision formation might improve PublicPST control, and it might reduce the tendency to push active EV actions to the maximum action boundary. It was not sufficient as final architecture evidence because the earlier comparison mixed a decision-architecture difference with an action-domain difference.

The current TD3 EV-GNN baseline explicitly uses the PublicPST charging-only mapping

$$a_i^{\text{base}} = \frac{a_{\max}}{2} [\tanh(z_i) + 1],$$

so both architectures operate in the same non-negative EV action domain. The 75k experiment also increases the design to **10 paired training seeds per architecture per scenario**, preserves the four formal PublicPST scenarios, and evaluates the selected `model.best` policy independently over 30 deterministic episodes with a separate infrastructure-diagnostic evaluation.

The primary study deliberately keeps TD3 fixed. Simultaneously replacing TD3 with SAC, PPO, TQC, or another optimiser would change both the optimisation algorithm and the policy architec-

ture, weakening attribution. Stronger model comparators are therefore treated as a later extension rather than part of this architecture-isolation experiment.

The 50k experiment is **motivation only**. Its observations are not pooled into the 75k effect estimates.

4 Formal experimental design

Design element	Formal setting
PublicPST scenarios	25CP, 100CP, 500CP, 1000CP
Architectures	2
Training seeds	10 per architecture per scenario
Total trained models	80
Training budget	75,000 environment steps
Scheduled evaluation cadence	every 5,000 steps
Scheduled evaluations per run	15
Episodes per scheduled evaluation	5
Checkpoint used for later final assessment	<code>model.best</code>
Independent final evaluation	30 deterministic episodes per trained model
Infrastructure diagnostic evaluation	30 episodes per trained model
Formal inferential unit	paired training seed; $n = 10$ per scenario

The formal experiment therefore contains $4 \times 2 \times 10 = 80$ independently trained models. The analysis first examines **training episodes and scheduled deterministic evaluations** to establish how optimisation progressed. The independent `model.best` evaluation and infrastructure diagnostics are then used for the final selected-policy comparison.

Training runs loaded: 80

Training-episode rows: 53,520

Scheduled-evaluation rows: 1,200

Scenarios: 25CP, 100CP, 500CP, 1000CP

Architectures: TD3 EV-GNN Baseline, Hierarchical TD3 EV-GNN

Scenario	CPs	Transformers	Ports/CS	Interval	Episode	V2G	Flexibility
25CP	25	3	1	15 min	112 steps / 28 h	Off	20
100CP	100	7	1	15 min	112 steps / 28 h	Off	5
500CP	500	35	1	15 min	112 steps / 28 h	Off	5
1000CP	1000	70	1	15 min	112 steps / 28 h	Off	5

CPs = charging points; Ports/CS = charging ports per charging station; Flexibility = the PublicPST power-setpoint flexibility parameter in the formal scenario configuration.

Shared training settings extracted from `run_args.yaml`: 75,000 steps; evaluation every 5,000 steps; 5 episodes per scheduled evaluation; batch size 64; learning rate 0.0003; discount 0.99; replay buffer 100,000; exploration noise 0.1; policy frequency 2; actor/critic GCN depth 3/3.

4.1 Cross-scale interpretation rule

The four configurations represent progressively larger formal PublicPST scenarios, but they are **not** a single-factor experiment in which only the number of charging points changes. Transformer count increases from **3 -> 7 -> 35 -> 70**, and the 25CP configuration uses a **setpoint-flexibility value of 20**, whereas the 100CP, 500CP, and 1000CP configurations use **5**. Cross-scale results should therefore be interpreted as robustness across progressively larger formal scenarios, not as a pure causal effect of charging-point count alone.

5 Three evidence layers

The final notebook separates three data-generating processes because they answer different questions:

1. **Training episodes (train_episode)** describe what happened during optimisation while exploration and parameter updates were occurring. They are noisy and correlated within a run.
2. **Scheduled deterministic evaluations (evaluation)** occur every 5,000 steps and provide a cleaner view of how policy quality evolved across the 75k budget.
3. **Independent model.best evaluation and infrastructure diagnostics** estimate selected-policy performance and operational mechanisms.

The training section therefore asks two questions before final-policy performance is considered: **what happened during optimisation?** and **how did deterministic policy quality evolve over the training budget?**

6 Training and scheduled-evaluation behaviour

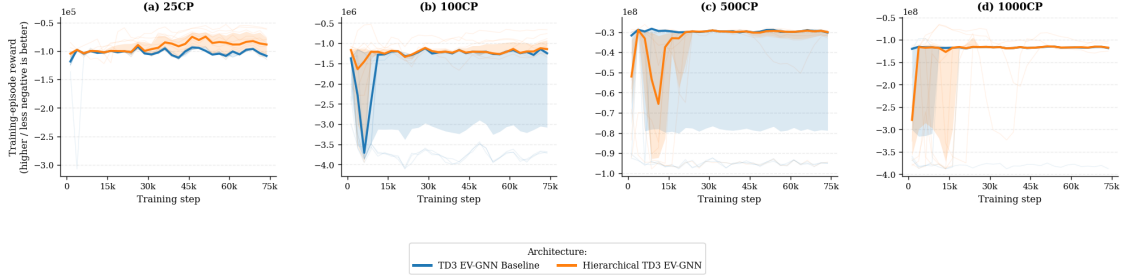
6.1 Figure 2 - Training optimisation behaviour

Training-episode reward describes what the actor experienced **while exploration and parameter updates were still taking place**. It is therefore noisier than deterministic evaluation reward, but it is useful for showing whether learning was stable, whether individual seeds underwent severe deterioration, and whether improvement occurred early or continued through the 75,000-step budget.

Figure 2 uses all training-episode observations. Each run is summarised in 2,500-step bins so that the full optimisation history remains visible without allowing episode-to-episode noise to dominate the figure. Thin lines represent the ten independently trained seeds for each architecture, the bold line gives the cross-seed median, and the shaded band gives the interquartile range.

How to read the figure. Moving upward means training reward is becoming less negative and therefore better under the PublicPST reward objective. A narrow band indicates that the ten seeds behave similarly; a broad band or isolated thin trajectory indicates stronger seed-to-seed variability. These are descriptive optimisation patterns, not the final selected-policy result.

Figure 2. Training optimisation behaviour across four formal PublicPST scenarios



Scenario	Baseline late-training median	Hierarchy late-training median	Hierarchy - baseline	Relative difference
25CP	-102.6k	-85.2k	17.4k	+17.0%
100CP	-1.2M	-1.2M	71.2k	+5.8%
500CP	-29.5M	-29.5M	1.4k	+0.0%
1000CP	-116.8M	-116.7M	23.1k	+0.0%

Observed optimisation pattern. Across the final 15,000 training steps, the hierarchical architecture has the higher cross-seed median training reward in **4 of 4** formal scenarios. The table reports the scale-specific magnitude. These values describe optimisation behaviour only; training episodes are correlated within runs and should not be treated as independent final-performance replicates.

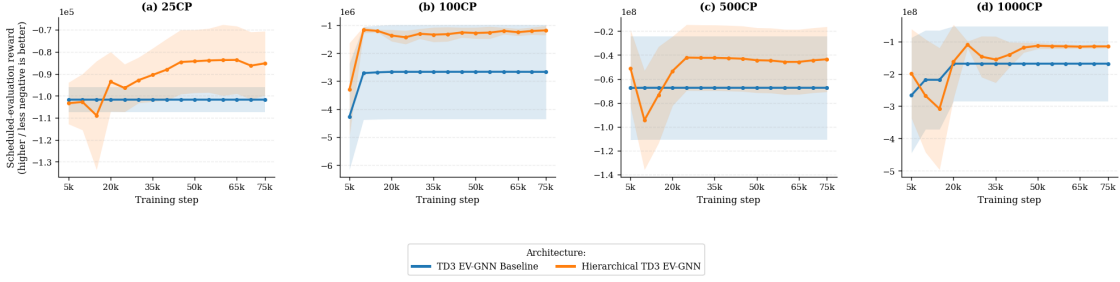
6.2 Figure 3A - Scheduled-evaluation learning curves

Training-episode reward is affected by exploration and ongoing parameter updates. Scheduled deterministic evaluation provides a cleaner view of how the policy itself changed over training. Every 5,000 steps, each run is evaluated for five deterministic episodes; Figure 3A then summarises the resulting evaluation reward across the ten independently trained seeds.

For each checkpoint, the solid line is the **mean across ten training seeds** and the shaded region is a **95% confidence interval across those seeds**. Higher / less-negative reward means better PublicPST objective performance.

How to read the figure. The vertical position shows policy quality at a particular checkpoint; the slope shows whether that quality improved or deteriorated as training continued; the width of the confidence band shows how consistently the ten independent training seeds behaved. Figure 3A should therefore be used to judge the timing and stability of learning, not to replace the independent `model.best` evaluation used for the final architecture comparison.

Figure 3A. Mean scheduled-evaluation learning curves with 95% seed-level confidence intervals



Scheduled evaluation at 75k training steps (mean [95% seed-level CI]; this is not the independent `model.best` eval30):

Scenario	TD3 EV-GNN Baseline	Hierarchical TD3 EV-GNN	Hierarchy - baseline	Relative difference
25CP	-101.7k [-107.4k, -96.0k]	-85.2k [-99.8k, -70.6k]	16.5k	+16.2%
100CP	-2.7M [-4.3M, -984.0k]	-1.2M [-1.4M, -1.0M]	1.5M	+55.6%
500CP	-67.4M [-110.7M, -24.2M]	-43.6M [-70.8M, -16.3M]	23.9M	+35.4%
1000CP	-168.6M [-284.5M, -52.7M]	-115.1M [-119.9M, -110.4M]	53.4M	+31.7%

Observed scheduled-evaluation pattern. At the final 75k checkpoint, the hierarchical architecture has the higher mean scheduled-evaluation reward in **4 of 4** scenarios. The scale-specific confidence intervals above describe uncertainty across the ten training seeds. They should not be substituted for the paired final-policy inference performed on `model.best` in Stage 2.

6.3 Figure 3B - Joint distribution of scheduled-evaluation performance

A mean learning curve can hide how widely the individual seed-checkpoint observations are distributed. Figure 3B therefore shows the **joint distribution of training time and scheduled-evaluation reward** for each architecture. The layout is adapted from the joint-density presentation used in Fig. 4a of Orfanoudakis et al. (2025), but the variables are chosen for the present learning question.

For each PublicPST scenario:

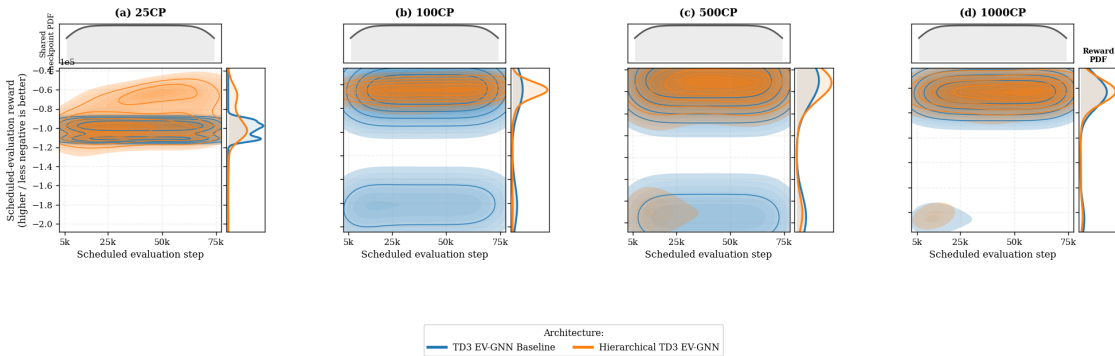
- the **central filled density** shows where the $10 \text{ seeds} \times 15 \text{ scheduled checkpoints}$ are concentrated;
- the **x-axis** is the scheduled checkpoint from 5k to 75k training steps;
- the **y-axis** is scheduled-evaluation reward;
- the **right marginal PDF** summarises the reward distribution after the training-time dimension is collapsed;

- the **top marginal PDF** represents the common checkpoint schedule. Because both architectures are evaluated at exactly the same checkpoints, this marginal is shared by design rather than being a performance result.

The four scenarios are plotted separately because their raw reward magnitudes differ substantially. Each architecture contributes 150 seed-checkpoint observations per scenario, but these are repeated measurements from ten trained seeds. Figure 3B is therefore **descriptive learning evidence**, not an inferential sample of $n = 150$.

Reference for the visual form: Orfanoudakis et al. (2025), Fig. 4a, <https://doi.org/10.1038/s44172-025-00457-8>. The accompanying EV-GNN repository implements its joint-density figure with filled KDE regions and marginal density curves in `Results_Analysis/generalization_eda.py`.

Figure 3B. Joint distribution of scheduled-evaluation performance over the 75,000-step learning process



How to read Figure 3B. Focus first on the vertical position of the filled density. A body concentrated higher on the y-axis means that more scheduled evaluations occurred at less-negative, and therefore better, PublicPST reward values. The width and shape of the body show how performance varies across training checkpoints and independent seeds. A compact body indicates that observations are concentrated in a narrower region; a broader or multi-region body indicates greater variation.

The **right marginal PDF** is the quickest way to compare the overall reward distributions within a scenario. If the hierarchical marginal is shifted upward relative to the baseline marginal, its scheduled evaluations are more often concentrated at better reward values. The **top marginal is deliberately neutral** because both architectures use the same 5k-to-75k evaluation schedule; it provides the joint-density layout without suggesting that checkpoint frequency differs between models.

Colour intensity represents **local observation density within an architecture**, not the magnitude of reward improvement. Figure 3B should therefore be read together with Figure 3A: Figure 3A shows the checkpoint-wise cross-seed mean and uncertainty, while Figure 3B shows where all scheduled seed-checkpoint observations are concentrated. Neither figure replaces the independent paired `model.best` comparison used for the final architecture result.

Observed joint-density pattern. Descriptively, the hierarchical architecture is concentrated in the higher, less-negative reward region at 25CP, while the baseline occupies a lower reward

band. At 100CP, the baseline shows a pronounced secondary low-reward density region, whereas the hierarchical observations are concentrated mainly in the higher reward band. The same broad pattern is visible at 500CP and 1000CP: the principal density regions overlap to some extent, but the baseline additionally contains distinct lower-reward mass that is much less prominent for the hierarchical architecture. This is consistent with the larger seed-to-seed dispersion visible in Figure 3A and with the early checkpoint selection observed for many baseline seeds. These density differences are **descriptive evidence about the learning process**; they do not establish the final architecture effect, which is assessed from paired independent `model.best` evaluations in Stage 2.

6.3.1 Checkpoint-selection robustness

Scenario	Architecture	Median best checkpoint	IQR	Best at 5k	Best $\leq 25k$	Best $> 50k$
25CP	Baseline	5k	5k-5k	10/10	10/10	0/10
25CP	Hierarchical	60k	47k-68k	2/10	2/10	7/10
100CP	Baseline	5k	5k-10k	6/10	9/10	1/10
100CP	Hierarchical	30k	15k-55k	0/10	5/10	4/10
500CP	Baseline	5k	5k-5k	8/10	10/10	0/10
500CP	Hierarchical	25k	8k-50k	3/10	6/10	3/10
1000CP	Baseline	5k	5k-8k	7/10	10/10	0/10
1000CP	Hierarchical	10k	5k-32k	4/10	7/10	2/10

Observed checkpoint-selection pattern. **25CP:** baseline 10/10 at 5k versus hierarchy 2/10; median best checkpoint 5k versus 60k. **100CP:** baseline 6/10 at 5k versus hierarchy 0/10; median best checkpoint 5k versus 30k. **500CP:** baseline 8/10 at 5k versus hierarchy 3/10; median best checkpoint 5k versus 25k. **1000CP:** baseline 7/10 at 5k versus hierarchy 4/10; median best checkpoint 5k versus 10k. A later selected checkpoint is not automatically a better final controller. It means only that the best scheduled evaluation for that seed was reached later in the training budget. Conversely, repeated selection of the first 5k checkpoint shows that later scheduled evaluations did not exceed the early best value for those seeds.

7 Final selected-checkpoint performance

The training section described optimisation and checkpoint selection. The evidence layer now changes: every result below uses the independently evaluated `model.best` policy.

For each trained seed s , the 30 deterministic evaluation episodes are first averaged,

$$\bar{y}_s = \frac{1}{30} \sum_{e=1}^{30} y_{s,e},$$

and architecture effects are then compared across the **10 paired training seeds**. The inferential unit is therefore $n = 10$ per formal scenario. Episode-level observations are still shown in selected figures because they reveal within-policy variability, but they are **descriptive background observations rather than additional independent training replicates**.

The main selected-policy analysis keeps four questions separate:

- **Episode reward — control efficacy.** Higher, less-negative reward is better under the PublicPST objective.
- **Power-tracker violation — tracking reliability.** Lower means less violation of the permitted tracking region.
- **Maximum active-EV action fraction — decision behaviour.** Lower means less upper-bound saturation, but this is not independently equivalent to better control.
- **Transformer overload — infrastructure consequence.** Lower is operationally preferable.

EV-service outcomes are examined separately so that a control improvement is not described as operationally better if it is obtained by materially reducing charging service.

Independent eval30 episodes: 2,400

Infrastructure-diagnostic episodes: 2,400

Seed-level policies: 80

Paired architecture comparisons: 40

7.1 Figure 4 - Selected-policy performance across control, decision and infrastructure dimensions

Figure 4 compares the **TD3 EV-GNN Baseline** directly with the **Hierarchical TD3 EV-GNN** under each formal scenario.

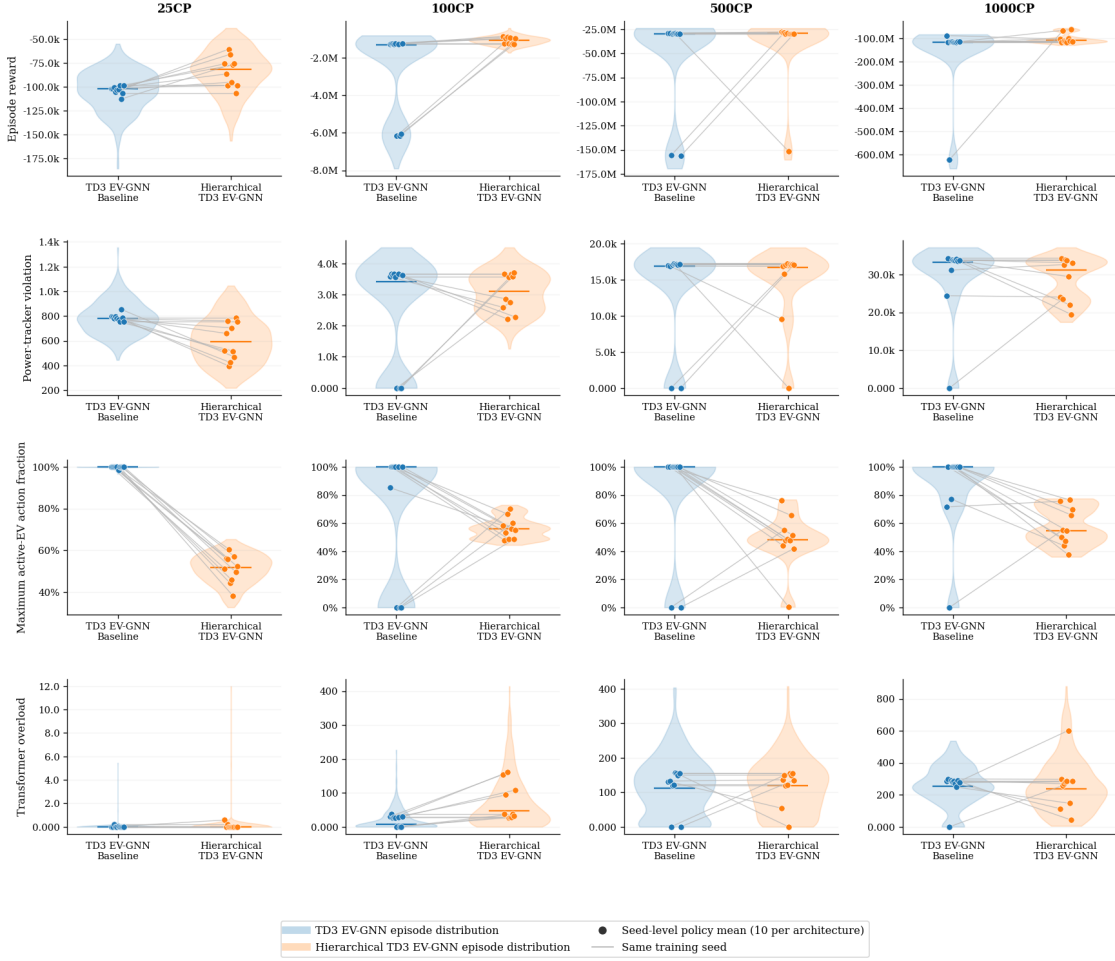
Each panel contains two evidence layers:

- the **light violin** contains all 300 independent-evaluation episode outcomes for one architecture and scenario and therefore shows descriptive within-policy variability;
- the **large coloured circles** are the 10 independently trained seed-level policy means used in formal paired inference;
- the **thin grey connector** joins the baseline and hierarchical result from the same training seed.

The architecture names are written explicitly on the horizontal axis. Each scenario deliberately keeps its **own vertical scale**, because raw reward, tracking-violation and overload magnitudes change substantially with scenario size. The tick labels use compact **k/M/B** notation rather than scientific-notation multipliers. The correct comparison is **between the two architectures within the same scenario**, not the absolute vertical position of different scenarios.

- **Episode reward:** higher / less negative is better.
- **Power-tracker violation:** lower is better.
- **Maximum active-EV action fraction:** lower means less upper-bound saturation; this is a behavioural change rather than an unconditional utility improvement.
- **Transformer overload:** lower is operationally preferable.

Figure 4. Selected-policy performance across control, decision and infrastructure dimensions



Absolute selected-policy outcomes (10 trained-policy seed means per architecture):

Scenario	Outcome	TD3 EV-GNN mean $\pm$ SD	Hierarchical TD3 EV-GNN mean $\pm$ SD	Change under hierarchy	Preferred direction	Result
25CP	Episode reward	-103.5k $\pm$ 4.2k	-84.3k $\pm$ 15.4k	19.2k (+18.5%)	Higher / less negative	Hierarchy favourable
25CP	Power-tracker violation	786 $\pm$ 28.8	598 $\pm$ 149	-187 under hierarchy (-23.8%)	Lower	Hierarchy favourable
25CP	Maximum active-EV action fraction	99.9% $\pm$ 0.5 pp	51.0% $\pm$ 6.8 pp	-48.8 percentage points	Context-dependent	Less upper-bound saturation

Scenario	Outcome	TD3 EV-GNN mean $\pm$ SD	Hierarchical TD3 EV-GNN mean $\pm$ SD	Change under hierarchy	Preferred direction	Result
25CP	Transformer overload	0.023 $\pm$ 0.058	0.075 $\pm$ 0.183	0.053 under hierarchy	Lower	Baseline favourable
100CP	Episode reward	-2.7M $\pm$ 2.4M	-1.1M $\pm$ 177.2k	1.6M (+59.9%)	Higher / less negative	Hierarchy favourable
100CP	Power- tracker violation	2.5k $\pm$ 1.7k	3.1k $\pm$ 608	549 under hierarchy (+21.7%)	Lower	Baseline favourable
100CP	Maximum active-EV action fraction	68.5% $\pm$ 47.5 pp	56.4% $\pm$ 7.6 pp	-12.1 percentage points	Context- dependent	Less upper- bound saturation
100CP	Transformer overload	20.6 $\pm$ 14.6	70.6 $\pm$ 54.2	50.0 under hierarchy	Lower	Baseline favourable
500CP	Episode reward	-54.8M $\pm$ 53.4M	-41.5M $\pm$ 38.7M	13.3M (+24.3%)	Higher / less negative	Hierarchy favourable
500CP	Power- tracker violation	13.7k $\pm$ 7.2k	14.5k $\pm$ 5.6k	831 under hierarchy (+6.1%)	Lower	Baseline favourable
500CP	Maximum active-EV action fraction	80.0% $\pm$ 42.2 pp	47.8% $\pm$ 19.6 pp	-32.2 percentage points	Context- dependent	Less upper- bound saturation
500CP	Transformer overload	112 $\pm$ 60.6	117 $\pm$ 50.9	5.3 under hierarchy	Lower	Baseline favourable
1000CP	Episode reward	-164.1M $\pm$ 160.8M	-102.5M $\pm$ 20.6M	61.6M (+37.5%)	Higher / less negative	Hierarchy favourable
1000CP	Power- tracker violation	29.3k $\pm$ 10.7k	28.6k $\pm$ 5.7k	-672 under hierarchy (-2.3%)	Lower	Hierarchy favourable
1000CP	Maximum active-EV action fraction	84.9% $\pm$ 31.7 pp	57.5% $\pm$ 13.6 pp	-27.3 percentage points	Context- dependent	Less upper- bound saturation
1000CP	Transformer overload	252 $\pm$ 89.7	258 $\pm$ 150	5.6 under hierarchy	Lower	Baseline favourable

7.1.1 Observed selected-policy pattern

Figure 4 shows that the hierarchical architecture does **not** produce one uniformly favourable operational profile.

- **25CP**: the hierarchical policy has a substantially higher mean reward (about **18.5%** relative to the baseline mean magnitude), lower mean power-tracker violation (about **23.8%** lower), and roughly **48.8 percentage points** less maximum-action use. The trade-off is that mean transformer overload is higher.
- **100CP**: mean reward is markedly higher under the hierarchy, but mean power-tracker violation and transformer overload are both higher. This is therefore a control-performance result with an infrastructure/tracking trade-off rather than an unqualified improvement.
- **500CP**: the hierarchical mean reward is higher and maximum-action use is much lower, but reward variability is large and the mean violation/overload outcomes are slightly adverse. The raw distributions therefore remain important.
- **1000CP**: mean reward is higher, mean power-tracker violation is slightly lower, and maximum-action use is reduced, while mean transformer overload is slightly higher and considerably more variable.

The episode distributions in the background explain why seed-level means alone can look sparse, but they **do not** increase the inferential sample size. Whether the reward advantage is statistically robust across independently trained policies is examined next with the ten paired seeds.

8 Paired architecture effects and formal inference

Figure 5 shows the pairing without first transforming the data into an abstract effect axis. Each row is one independently trained seed. The blue point is that seed’s **model.best** reward under the **TD3 EV-GNN Baseline**, the orange point is the reward under the **Hierarchical TD3 EV-GNN**, and the grey segment joins the same seed pair.

Because reward is negative in these experiments, a point farther to the **right** is better: it is higher / less negative. The figure therefore answers a direct question:

For each independently trained seed, which architecture produced the better selected-policy reward?

The mean paired improvement and its 95% confidence interval are reported in Table 2 rather than mixed into the figure legend.

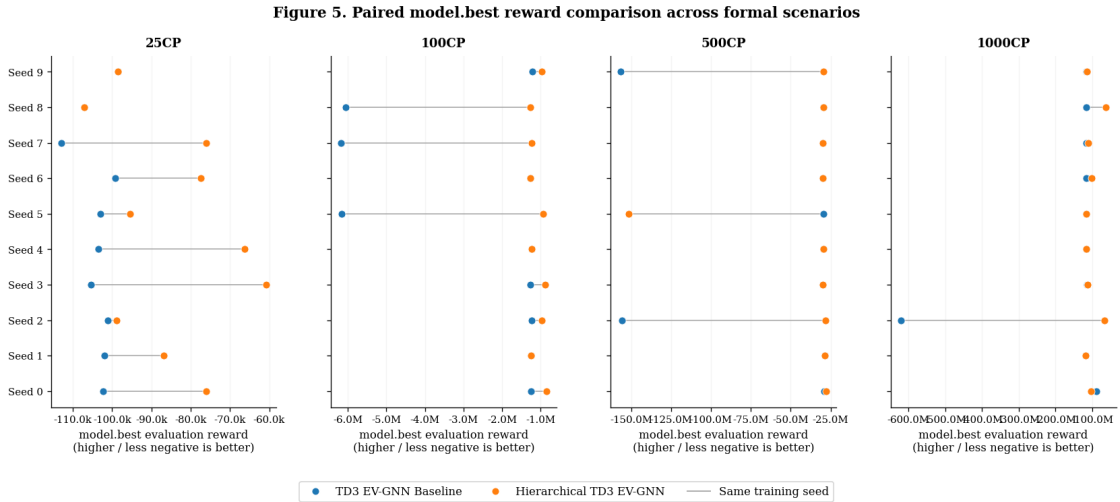


Table 2. Primary paired reward evidence across the four formal scenarios.

Scenario	Average reward improvement under hierarchy	95% CI of paired improvement	Seed pairs with higher reward under hierarchy	Holm-adjusted p-value	Evidence conclusion
25CP	19.2k	[7.2k, 31.2k]	9 of 10	0.022	Supported reward improvement
100CP	1.6M	[-39.3k, 3.3M]	9 of 10	0.164	Strong favourable direction; mean CI includes zero
500CP	13.3M	[-37.6M, 64.2M]	6 of 10	0.583	Mixed and heterogeneous
1000CP	61.6M	[-62.7M, 185.9M]	7 of 10	0.583	Positive direction; mean CI includes zero

8.1 Formal reward interpretation

How to read the 25CP row. The hierarchical policy achieved an average paired reward improvement of about **19.2k** relative to the same ten baseline seeds. The 95% confidence interval is approximately **[+7.2k, +31.2k]**, **9 of 10** seed pairs have higher reward under the hierarchy, and the reward result remains below 0.05 after correcting the four-scale reward family with Holm (**adjusted** $p = 0.022$). The 25CP scenario therefore provides the strongest confirmatory evidence that infrastructure-aligned decision factorisation improves PublicPST control.

At **100CP**, **9 of 10** seed pairs again favour the hierarchy, but the mean paired confidence interval narrowly crosses zero. This is strong directional evidence, but not the same statistically resolved mean-effect result observed at 25CP.

At **500CP**, only **6 of 10** seed pairs favour the hierarchy and the paired effects are highly heterogeneous. At **1000CP**, **7 of 10** pairs favour the hierarchy and the mean effect is positive, but the confidence interval remains wide. The larger-scenario evidence therefore shows that the control advantage remains observable as scale increases, while its reproducibility becomes more sensitive to training seed and configuration.

The complete paired t-test, Wilcoxon, exact sign-flip and Cohen d_z details are retained in Appendix B so that the main table remains readable.

9 Decision mechanism: action composition

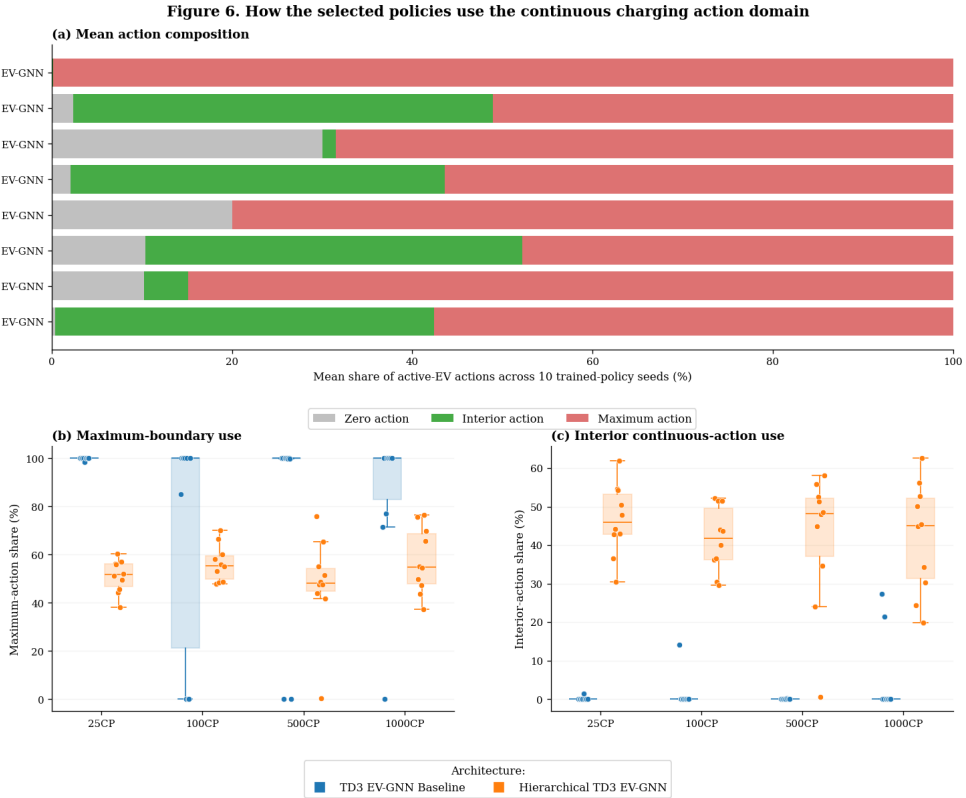
For every active EV, the selected policy places the action in one of three mutually exclusive regions:

$$f_0 + f_{\text{interior}} + f_{\text{max}} = 1.$$

Figure 6 replaces the ternary plot with a more direct composition view.

- **Figure 6A** shows the mean action composition across the 10 trained-policy seeds. Every bar totals 100%, so the reader can see how much decision mass lies at zero action, inside the continuous action range, or at the maximum boundary.
- **Figure 6B** shows the 10 seed-level distributions for the two components most relevant to the observed mechanism: maximum-action use and interior-action use.

The purpose is not to claim that interior action is always better. The question is whether the hierarchy changes a strongly boundary-dominated policy into a more graded continuous controller, and whether that change coexists with acceptable tracking and service outcomes.



9.1 Observed action-composition shift

The hierarchical policy moves away from maximum-boundary dominated control in **all four** formal scenarios. Relative to the paired baseline policies, mean maximum-action use falls by approximately **48.8 percentage points at 25CP, 12.1 at 100CP, 32.2 at 500CP, and 27.3 at 1000CP**.

The destination of that mass is also visible. Interior-action use increases by about **46.4, 40.1, 41.8, and 37.2 percentage points**, respectively. At 25CP, zero-action use increases slightly, whereas at 100CP, 500CP and 1000CP it decreases. The mechanism is therefore not simply “less

charging”: the hierarchy generally replaces a substantial part of maximum-boundary behaviour with graded interior actions.

This is one of the clearest architecture-level behavioural differences in the study. It remains a mechanism result rather than a standalone performance claim; Figures 4, 7 and the service outcomes determine whether that behavioural shift is operationally beneficial.

10 Infrastructure allocation mechanism

The hierarchy introduces explicit transformer- and charger-level allocation weights, so the diagnostics also examine **how concentrated positive charging authority is within a scenario**.

For positive allocation shares s_i ,

$$HHI = \sum_i s_i^2.$$

Higher HHI means that positive allocation is more concentrated; lower HHI means it is more distributed. The Gini coefficient describes allocation inequality in a similar descriptive sense. **Neither metric is inherently good or bad.** Concentration can be rational when demand or available capacity is uneven.

Raw HHI is entity-count sensitive, so the main comparison is horizontal: baseline versus hierarchy **within the same formal scenario**. Detailed HHI/Gini episode distributions are retained in Appendix D.

Positive-charge allocation concentration (seed-level means; compare architectures within each scenario):

Scenario	TD3 EV-GNN trans- former HHI	Hierarchical TD3 EV-GNN trans- former HHI	Change under hierarchy (trans- former HHI)	TD3 EV-GNN charger HHI	Hierarchical TD3 EV-GNN charger HHI	Change under hierarchy (charger HHI)
25CP	0.387	0.421	+0.033	0.115	0.149	+0.034
100CP	0.175	0.195	+0.020	0.046	0.063	+0.017
500CP	0.146	0.125	-0.021	0.088	0.093	+0.005
1000CP	0.021	0.021	+0.000	0.007	0.007	+0.000

10.1 Observed allocation pattern

Within **25CP**, the hierarchical actor is more concentrated at both infrastructure levels: transformer HHI rises from about **0.387 to 0.421** and charger HHI from **0.115 to 0.149**. The same direction appears at **100CP** (transformer **0.175 → 0.195**, charger **0.046 → 0.063**).

At **500CP**, the mechanism is mixed: transformer allocation becomes less concentrated (**0.146 → 0.125**), while charger HHI changes only slightly upward (**0.088 → 0.093**). At **1000CP**, the mean HHI values are effectively unchanged at the reported precision.

These results do not establish that more concentrated or more distributed allocation is preferable. They show that the hierarchical architecture changes **where decision authority is concentrated**, and that this change itself is scale-dependent.

11 Operational trade-offs

A higher PublicPST reward does not automatically imply lower infrastructure stress. Figure 7 therefore compares the two architectures **within the same paired training seed**.

For power-tracker violation V and transformer overload O ,

$$\Delta_{\text{tracking}} = V^{\text{TD3 EV-GNN}} - V^{\text{hierarchical}},$$

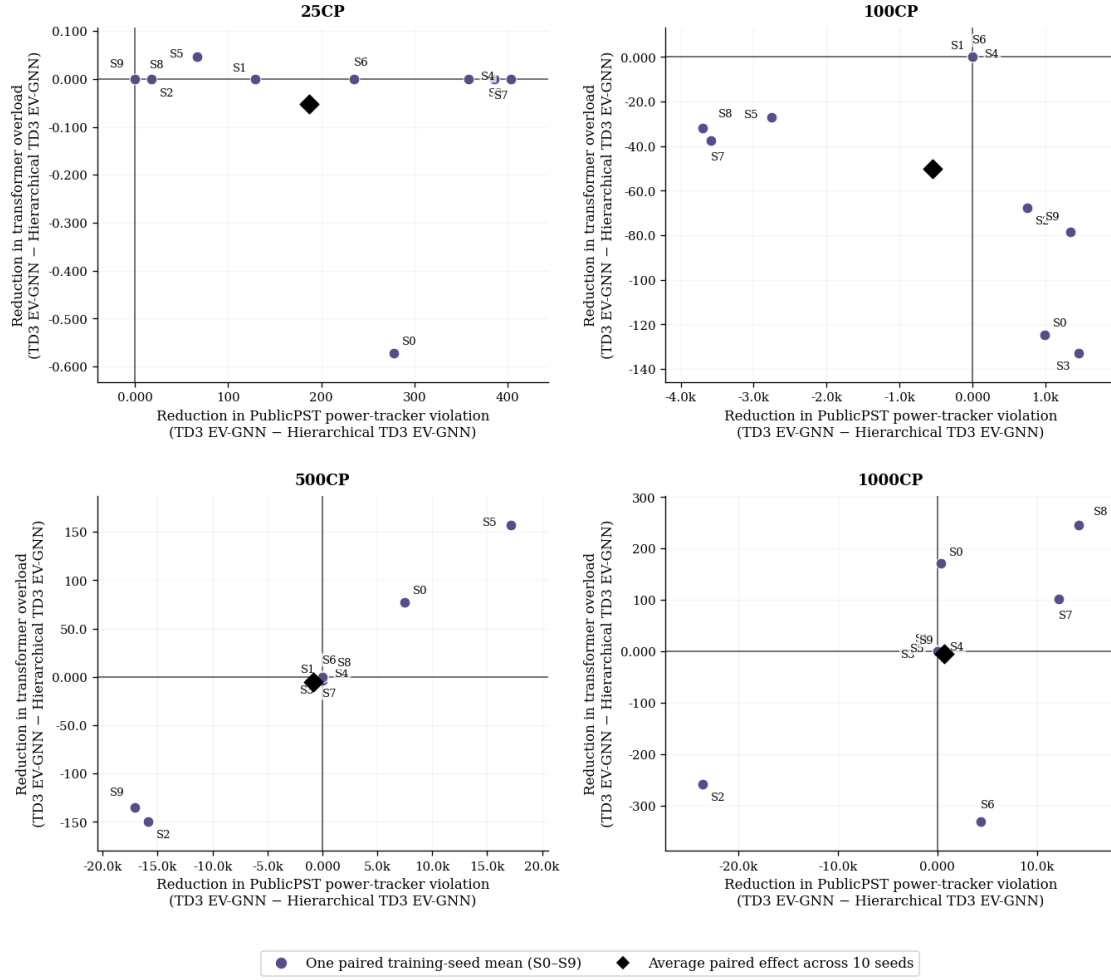
$$\Delta_{\text{overload}} = O^{\text{TD3 EV-GNN}} - O^{\text{hierarchical}}.$$

Each point in Figure 7 is one paired training-seed mean:

- $x > 0$: the hierarchical policy has **fewer PublicPST power-tracker violations**;
- $x < 0$: the TD3 EV-GNN baseline tracks better;
- $y > 0$: the hierarchical policy has **lower transformer overload**;
- $y < 0$: the TD3 EV-GNN baseline has lower overload.

Thus the upper-right quadrant is the clearest operational win for the hierarchy: better tracking **and** lower transformer stress. The lower-right quadrant means better tracking but higher overload. The four scenarios are shown as large 2×2 panels so the ten paired seeds remain readable.

Figure 7. Does better PublicPST tracking coincide with lower transformer stress?



EV-service changes accompanying the control result:

Scenario	Mean change in user satisfaction under hierarchy	Seed pairs with higher satisfaction under hierarchy	Mean change in total energy delivered under hierarchy	Seed pairs with more energy delivered under hierarchy
25CP	-0.19 percentage points	0 of 10	-24.6	1 of 10
100CP	+6.55 percentage points	3 of 10	1.4k	3 of 10
500CP	+2.05 percentage points	2 of 10	2.2k	3 of 10
1000CP	+1.94 percentage points	2 of 10	4.1k	2 of 10

11.1 Observed tracking–infrastructure trade-off

The trade-off is clearly scale-dependent.

- **25CP:** mean tracking improves, but mean transformer overload increases. The control gain therefore comes with a measurable infrastructure cost in this scenario.
- **100CP:** the mean violation and overload differences both favour the baseline, even though several individual seeds improve tracking under the hierarchy. A small number of adverse seeds materially affect the mean.
- **500CP:** the seed cloud spans both sides of the tracking and overload axes. This is a genuinely mixed scenario rather than a clean operational win for either architecture.
- **1000CP:** mean power-tracker violation is slightly lower under the hierarchy, while mean overload is slightly higher. Importantly, **7/10** seeds actually show lower overload under the hierarchy, so the negative mean is driven by a minority of large adverse overload differences.

The EV-service table exposes the same mean-versus-consistency issue. Mean satisfaction is positive at 100CP, 500CP and 1000CP, but only **3/10**, **2/10** and **2/10** paired seeds favour the hierarchy, respectively. These means are therefore not evidence of a broadly shared service improvement.

12 Seed-direction consistency across outcomes

Effect magnitude and directional consistency answer different questions. Figure 8 deliberately ignores magnitude and asks only:

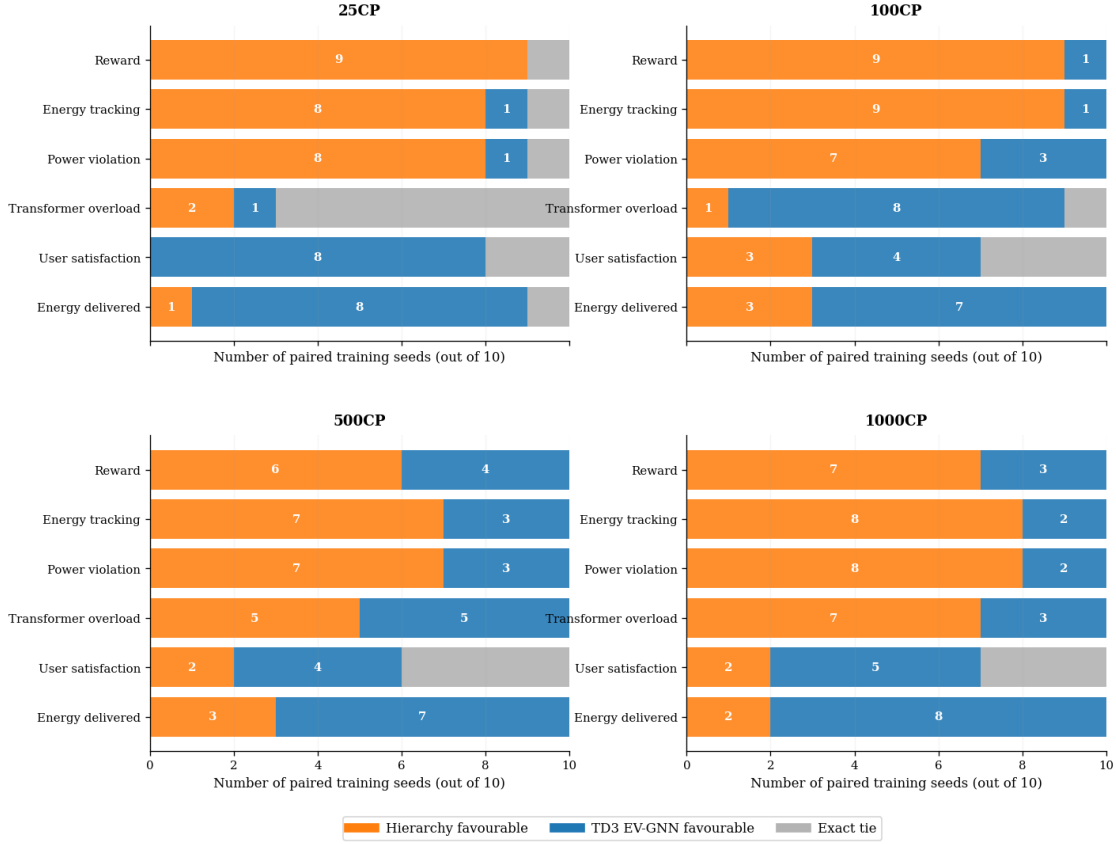
How often does the architecture effect repeat in the same direction across the 10 independently trained seed pairs?

Each horizontal bar totals ten seeds. The **orange segment** counts seed pairs in which the hierarchical architecture is favourable for that outcome; the **blue segment** counts seed pairs in which the TD3 EV-GNN baseline is favourable; grey would indicate exact ties.

This is a reproducibility view, not an effect-size plot. A 9/10 bar means that nine independently trained seed pairs move in the hierarchy-favourable direction; it does not mean that the nine effects have equal magnitude.

Grey indicates an **exact tie** in the seed-level metric estimate; it does not indicate missing data.

Figure 8. Directional consistency across the 10 paired training seeds



12.1 What Figure 8 says

At **25CP**, the control improvement is highly reproducible: reward, energy tracking and power-tracker violation favour the hierarchy in **9/10**, **8/10** and **8/10** paired seeds. That consistency does **not** extend to infrastructure or user service: only **2/10** seeds show lower overload and **0/10** show higher user satisfaction. The 25CP result is therefore a strong control gain with a clear operational trade-off.

At **100CP**, reward and energy tracking favour the hierarchy in **9/10** seeds, so the control direction is common even though the mean reward CI includes zero. Transformer overload, however, favours the hierarchy in only **1/10** seeds. The architecture therefore frequently improves the control objective without providing the same infrastructure benefit.

The **500CP** scenario is genuinely mixed: reward favours the hierarchy in **6/10** seeds, tracking measures in **7/10**, overload in **5/10**, and satisfaction in only **2/10**. No single directional claim represents this scenario well.

At **1000CP**, reward, energy tracking, power-tracker violation and overload favour the hierarchy in **7/10**, **8/10**, **8/10** and **7/10** seeds, respectively. User satisfaction favours the hierarchy in only **2/10** seeds. This is important because the mean satisfaction change is positive: a minority of larger gains raises the mean even though most paired seeds do not improve. Mean effects and seed

consistency therefore have to be read together.

13 Cross-scale synthesis and research findings

The four scenarios are progressively larger **configured PublicPST environments**, not a single-factor experiment in charging-point count. The synthesis therefore asks where the hierarchy is favourable, where evidence remains uncertain, and what operational trade-offs accompany the control result.

The table keeps checkpoint timing separate from robustness. A later best checkpoint only means that the best scheduled evaluation occurred later; it does not by itself prove more stable or better optimisation.

Table 3. Cross-scale research synthesis.

Scenario	Control evidence	Decision mechanism	Infrastructure evidence	EV-service evidence	Overall conclusion
25CP	Reward 19.2k; 95% CI [7.2k, 31.2k]; 9/10 seeds	Maximum-action share -48.8 pp; interior-action share +46.4 pp	Lower overload in 2/10 paired seeds	Higher satisfaction in 0/10 paired seeds	Supported control gain with a clear graded-action mechanism; infrastructure and service trade-offs remain.
100CP	Reward 1.6M; 95% CI [-39.3k, 3.3M]; 9/10 seeds	Maximum-action share -12.1 pp; interior-action share +40.1 pp	Lower overload in 1/10 paired seeds	Higher satisfaction in 3/10 paired seeds	Strong control direction but unresolved mean reward effect; overload is mostly baseline-favourable.
500CP	Reward 13.3M; 95% CI [-37.6M, 64.2M]; 6/10 seeds	Maximum-action share -32.2 pp; interior-action share +41.8 pp	Lower overload in 5/10 paired seeds	Higher satisfaction in 2/10 paired seeds	Mixed efficacy and infrastructure evidence despite a clear shift toward graded interior actions.

Scenario	Control evidence	Decision mechanism	Infrastructure evidence	EV-service evidence	Overall conclusion
1000CP	Reward 61.6M; 95% CI [-62.7M, 185.9M]; 7/10 seeds	Maximum- action share -27.3 pp; interior-action share +37.2 pp	Lower overload in 7/10 paired seeds	Higher satisfaction in 2/10 paired seeds	Control and overload directions are often hierarchy- favourable, but uncertainty and service inconsistency remain.

13.1 Results and interpretation

13.1.1 Primary improvement: a more graded continuous-control mechanism

The most reproducible architecture-level result is the action mechanism. Across **all four** formal PublicPST scenarios, the Hierarchical TD3 EV-GNN substantially reduces maximum-boundary charging and shifts decisions into the interior of the continuous action range. Mean maximum-action use falls by approximately **48.8, 12.1, 32.2 and 27.3 percentage points** from 25CP to 1000CP, while interior-action use increases by approximately **46.4, 40.1, 41.8 and 37.2 percentage points**.

This is stronger than saying that the architecture merely “changes behaviour”. The hierarchy consistently replaces a strongly boundary-dominated decision pattern with substantially more graded continuous control. Because TD3, the graph state, reward family, PublicPST environment and charging-only $[0, 1]$ action interface are held fixed, this behavioural difference is attributable to the change in **decision formation** rather than a change of reinforcement-learning algorithm or action domain.

13.1.2 Control efficacy across formal scenarios

At **25CP**, the behavioural change translates into the strongest control-efficacy result: mean selected-policy reward improves by about **19.2k** (approximately **18.5%** relative to the baseline mean magnitude), the 95% paired CI is approximately **[+7.2k, +31.2k]**, **9/10** seeds favour the hierarchy, and the reward result remains supported after Holm correction (**adjusted $p = 0.022$**). Mean power-tracker violation also falls from approximately **785.8 to 598.5**.

At the larger scenarios, the control advantage remains visible rather than disappearing. Mean selected-policy reward favours the hierarchy by approximately **59.9% at 100CP, 24.3% at 500CP and 37.5% at 1000CP** relative to the corresponding baseline mean magnitudes. However, the seed-to-seed uncertainty is much larger. Reward favours the hierarchy in **9/10** paired seeds at 100CP, **6/10** at 500CP and **7/10** at 1000CP, while the paired mean CIs still include zero. The evidence therefore supports a control advantage that remains observable as scenario size increases, but whose reproducibility becomes more sensitive to seed and configuration.

13.1.3 Why the trade-offs matter

The hierarchy does **not** provide a universal operational win in its present form. Transformer overload is not consistently lower, and user-satisfaction means can disagree with the majority seed direction. This means that improved fleet-level tracking cannot be treated as equivalent to improved infrastructure operation or improved user service.

The research value is therefore not that every stakeholder metric is already superior. The value is that explicit infrastructure-aligned decision factorisation produces a repeatable and interpretable control mechanism, demonstrates that this mechanism can materially improve PublicPST control, and exposes the remaining allocation trade-offs in a form that can now be isolated by architectural ablation.

14 Practical interpretation

Charging Point Operator. The hierarchical architecture can deliver a materially better PublicPST response, with the clearest statistically supported improvement at 25CP and favourable reward directions at larger scenarios. Its most consistent operational characteristic is that it relies much less on all-at-maximum EV commands and uses the continuous action range more gradually. For a CPO, this is a more graded and less boundary-dominated decision mechanism than a policy that frequently saturates active EV actions at the upper boundary.

Distribution infrastructure. The same control improvement does not automatically reduce transformer stress. The overload results are scale- and seed-dependent, and 25CP/100CP contain important adverse cases. The present hierarchy should therefore be interpreted as improving **how charging authority is formed**, not as having already solved infrastructure-aware allocation.

EV users. Better CPO tracking also does not guarantee better user service. Satisfaction is not consistently hierarchy-favourable across independently trained seeds. This separates the system-level control benefit from the user-service objective and identifies a second design dimension that future reward or allocation refinements may need to address.

Research value relative to the baseline. The TD3 EV-GNN baseline already represents the physical network with a graph, but it forms final EV actions directly. The hierarchical architecture uses the same topology to structure **decision authority** through transformer allocation, charger allocation and EV gating. The 75k evidence shows that this is not a cosmetic architectural change: it consistently alters action-space use and can produce substantial control gains. It also creates decision layers that can be observed, removed and redesigned, providing an architectural attribution pathway that the direct-action baseline does not offer.

15 Conclusion

15.1 What did the study demonstrate?

This study provides evidence that explicitly aligning TD3 EV-GNN **decision formation** with the physical CPO–Transformer–Charger–EV hierarchy is a meaningful architectural improvement over direct EV-node action formation. The clearest and most reproducible effect is mechanistic: across all four formal PublicPST scenarios, the hierarchical actor substantially reduces upper-bound action saturation and shifts control toward graded interior actions. The physical topology

is therefore useful not only as a graph representation, but also as a structure for allocating decision authority.

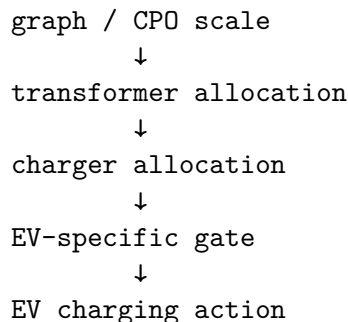
15.2 Where did the hierarchy outperform the baseline?

The strongest confirmatory evidence occurs at **25CP**. Selected-policy reward improves from approximately **−103.5k to −84.3k** (about **+18.5%**), mean power-tracker violation falls from about **785.8 to 598.5**, **9/10** paired seeds achieve higher reward, and the four-scale Holm-adjusted reward result remains supported at $p = 0.022$.

The control advantage does not disappear at larger scenarios. Mean selected-policy reward remains higher under the hierarchy at **100CP (+59.9%)**, **500CP (+24.3%)** and **1000CP (+37.5%)** relative to the corresponding baseline mean magnitudes. The direction is especially common at 100CP (**9/10** reward seed pairs) and remains present at 1000CP (**7/10**), while 500CP is genuinely mixed (**6/10**). These larger-scenario effects are less precisely resolved because seed variability becomes much larger, so they support a favourable control direction rather than the same confirmatory mean-effect claim established at 25CP.

15.3 Why is this valuable relative to the TD3 EV-GNN baseline?

The baseline uses the physical topology to learn graph representations but still forms final EV charging actions directly. The hierarchical policy goes further: it turns that topology into an explicit decision structure,



The four-scale 75k evidence shows that this structural change has measurable consequences. It consistently replaces boundary-dominated charging with more graded continuous control and can translate that mechanism into materially better PublicPST tracking. In addition, the explicit transformer, charger and EV decision levels can now be **observed, ablated, attributed and redesigned**. That provides a research lever that the direct-action baseline does not expose.

15.4 What trade-offs and risks remain?

The current full hierarchy does not yet provide uniformly favourable operational outcomes. Transformer overload is not consistently improved, EV-user satisfaction is not consistently hierarchy-favourable, and 500CP/1000CP contain substantial seed sensitivity and tail failures. Mean and majority-seed directions sometimes disagree, showing that a small number of large outcomes can dominate an average.

These results do not invalidate the hierarchical direction. They identify where the current factorisation still needs refinement: the architecture has improved the **control mechanism**, but it

does not yet guarantee that the resulting charging authority is allocated in a way that consistently reduces transformer stress or preserves EV-user service.

15.5 What experiment follows logically?

The next experiment should therefore determine **which hierarchical layer creates the benefit and which layer may create the cost**. The immediate comparison removes charger-level decision allocation while retaining transformer-level hierarchy. This tests whether the charger layer adds genuine control value or instead introduces unnecessary allocation constraints, infrastructure trade-offs or optimisation variability.

The resulting thesis position is therefore concrete: **infrastructure-aligned decision factorisation is a promising and empirically supported control direction; the next step is layer-wise attribution and refinement, not abandonment of the hierarchical approach.**

16 Claim boundaries, risks and next experiment

The current evidence supports conclusions for the four formal PublicPST configurations analysed here. It does not establish unseen-topology generalisation, unseen-demand generalisation, V2G performance, or superiority across reinforcement-learning algorithms.

16.1 Immediate architecture-attribution experiment

The next comparison is the **Transformer-to-EV Hierarchical TD3 EV-GNN with Charger-Level Decision Allocation Removed**:

Full hierarchy

CPD / graph scale

→ Transformer allocation

→ Charger allocation

→ EV gate

→ EV action

Charger-level allocation removed

CPD / graph scale

→ Transformer allocation

→ EV decision

The formal research question is:

Does explicit charger-level allocation provide additional control value beyond transformer-level hierarchical decision formation, or does it introduce unnecessary allocation constraints that contribute to the observed infrastructure, service and seed-variability trade-offs?

The comparison should retain the same four formal scenarios, 10 paired training seeds, 75,000-step budget, selected `model.best` eval30 protocol and infrastructure diagnostics. Six outcomes require particular attention:

1. **Reward and tracking:** does the control benefit remain after removing the charger layer?

2. **Maximum/interior action behaviour:** is saturation reduction caused by hierarchy generally, or specifically by charger allocation?
3. **Transformer overload:** do the adverse 25CP/100CP overload effects improve?
4. **EV service:** do satisfaction and energy-service outcomes become more consistent?
5. **Seed variability:** are the large 500CP/1000CP tail failures reduced?
6. **Checkpoint timing:** does the reduced architecture still require later training checkpoints to find its best policy?

16.1.1 Competing explanations

H1 — Charger-level allocation is beneficial. If removing it reduces reward/tracking performance or restores maximum-boundary saturation, charger-level factorisation contributes genuine control value.

H2 — Charger-level allocation is over-constraining. If the reduced hierarchy preserves reward/tracking and the graded-action mechanism while improving overload, service or seed variability, the charger layer is likely adding unnecessary constraint or optimisation difficulty.

H3 — Transformer-level allocation is the principal contributor. If the reduced hierarchy and full hierarchy are materially similar, the contribution should be narrowed to transformer-level hierarchical decision formation rather than claiming that the full three-level factorisation is necessary.

If both hierarchical variants remain highly variable, the next priority should be optimisation stability or factorisation design rather than adding additional hierarchy.

16.2 Later research

After architecture attribution, **TD3 FX-GNN** remains the preferred stronger external comparator because TD3 stays fixed while representation/policy design changes. If large-scale uncertainty remains unresolved, targeted additional seeds at 500CP/1000CP would be more informative than immediately expanding to a broad SAC/PPO/TQC benchmark. Unseen topology, unseen demand distributions and V2G should remain later research rather than being added to the current thesis scope.

17 Appendix A - Metric glossary

Metric	What it measures	Better direction	Operational meaning	Main caution
Episode reward	PublicPST objective outcome	Higher / less negative	Overall control efficacy under the simulator reward	Composite task objective; inspect underlying tracking/service metrics
Tracking error	Aggregate setpoint mismatch	Lower	Requested and realised charging power are closer	Exact simulator definition should be retained

Metric	What it measures	Better direction	Operational meaning	Main caution
Energy-tracking error	Accumulated energy mismatch	Lower	Better cumulative energy alignment	Distinct from instantaneous power violation
Power-tracker violation	Violation burden around the requested trajectory	Lower	More reliable adherence to the permitted tracking region	Must be interpreted with the configured tolerance
Zero-action fraction	Active-EV decisions at the lower boundary	Contextual	How often active EVs receive no charging command	High can be appropriate or pathological
Maximum-action fraction	Active-EV decisions at the upper boundary	Contextual	Degree of upper-bound saturation	Lower is not automatically better
Interior-action fraction	Active-EV actions away from both boundaries	Contextual	Use of graded continuous control	Useful only with acceptable tracking and service
Transformer overload	Aggregate transformer overload diagnostic	Lower	Lower infrastructure stress	Magnitude is scenario-dependent
HHI	Positive allocation concentration	Contextual	Concentrated vs distributed authority	Entity-count sensitive; not a performance score
Gini	Positive allocation inequality	Contextual	Unequal vs even allocation	Not a performance score
User satisfaction	Simulator-defined EV-user service outcome	Higher	Better charging-service experience	Simulator-defined measure
Total energy charged	Charging service delivered	Contextual / generally higher when demand exists	Amount of charging energy supplied	Must be interpreted relative to task demand
EVs served	Number of EV service outcomes observed	Contextual	Coverage of charging demand	Same evaluation seeds can make counts identical across policies

18 Appendix B - Statistical glossary

95% confidence interval. In this notebook the interval around a paired mean effect quantifies uncertainty across the ten independently trained paired seeds. An interval that crosses zero means the observed seed sample remains compatible with both positive and negative mean effects.

Paired t-test. The paired t-test evaluates whether the mean within-seed architecture difference is zero. The implementation uses `scipy.stats.ttest_rel`, matching the paired experimental design. Reference: https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.ttest_rel.html

Wilcoxon signed-rank test. This rank-based paired test is less directly tied to the mean than the paired t-test and provides a complementary robustness view when the seed-difference distribution is irregular.

Reference: <https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.wilcoxon.html>

Exact sign-flip test. With ten paired seeds, all $2^{10} = 1024$ possible sign assignments of the observed paired differences can be enumerated. The two-sided p-value is the fraction of sign assignments producing an absolute mean at least as large as the observed one.

Paired standardised effect d_z .

$$d_z = \frac{\bar{d}}{s_d},$$

where $\bar{d}$ is the mean paired difference and s_d is the standard deviation of the paired differences. The sign follows the notebook’s benefit orientation.

General effect-size reporting reference: Lakens (2013), <https://doi.org/10.3389/fpsyg.2013.00863>

Holm correction. The four primary **paired t-test p-values for reward** form one family because the same reward question is asked at 25CP, 100CP, 500CP, and 1000CP. Holm orders the four p-values and applies progressively less stringent thresholds. It controls family-wise false-positive risk; it is not a model-performance metric.

Reference: Holm (1979), <https://www.jstor.org/stable/4615733>

Supplementary Table B1. Complete paired reward statistics.

Scenario	Mean paired reward improvement	95% CI	Cohen dz	Seeds hierarchy > baseline	Paired t-test p	Wilcoxon p	Exact sign-flip p	Holm-adjusted paired-t p
25CP	19.2k	[7.2k, 31.2k]	1.14	9/10	0.006	0.004	0.004	0.022
100CP	1.6M	[-39.3k, 3.3M]	0.70	9/10	0.055	0.010	0.010	0.164
500CP	13.3M	[-37.6M, 64.2M]	0.19	6/10	0.568	0.432	0.332	0.583

Scenario	Mean paired reward im- prove- ment	95% CI	Cohen dz	Seeds hierar- chy > baseline	Paired t-test p	Wilcoxon p	Exact sign-flip p	Holm- adjusted paired-t p
1000CP	61.6M	[-62.7M, 185.9M]	0.35	7/10	0.291	0.084	0.133	0.583

19 Appendix C - Seed-level checkpoint stability maps

The main learning figures summarise performance across seeds. This appendix asks a narrower question that cannot be read reliably from an average curve:

For each independently trained seed, when was its best scheduled checkpoint reached, and what happened before and after that point?

The map therefore does **not** colour cells by raw reward. Instead, each row is normalised relative to that seed’s own best scheduled-evaluation reward:

$$\text{gap}_{s,t} = 100 \times \frac{R_s^{\text{best}} - R_{s,t}}{|R_s^{\text{best}}|}.$$

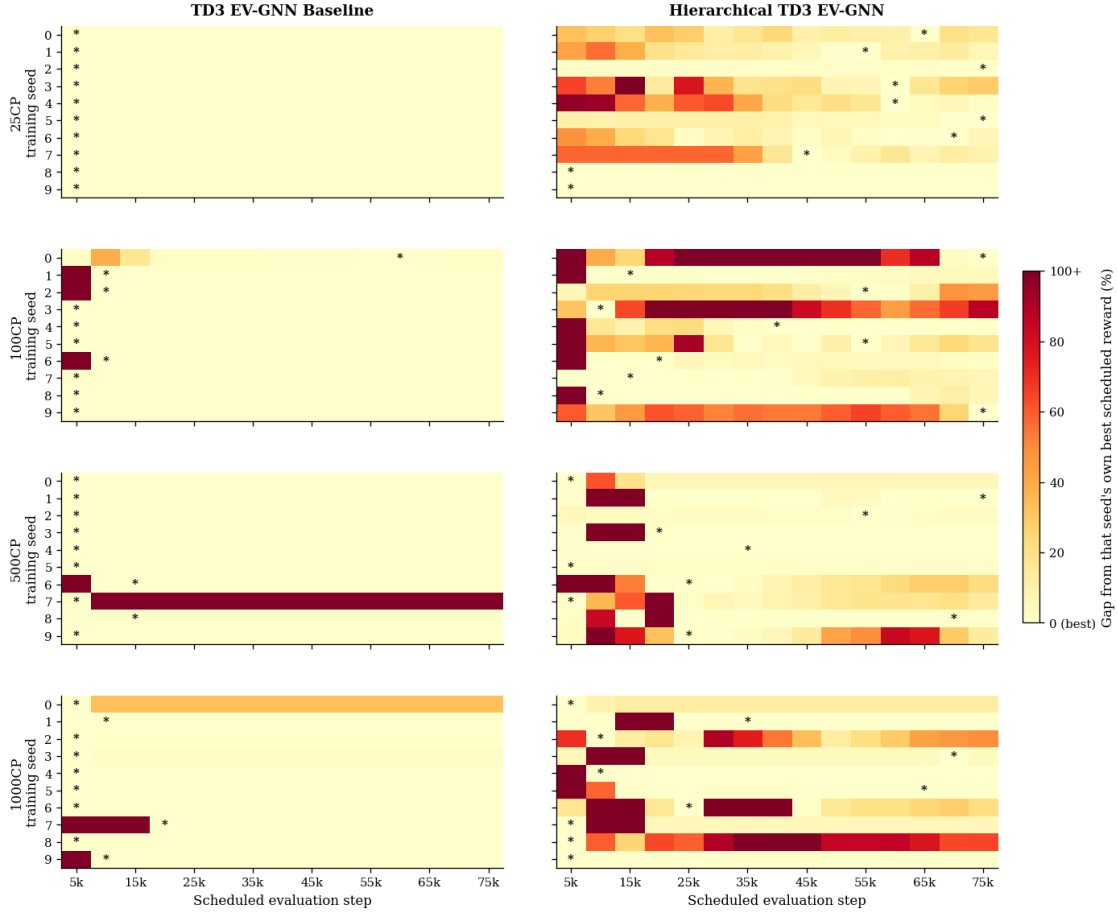
A value of **0%** means that checkpoint is tied with the best scheduled evaluation observed for that seed. A value of **20%** means the checkpoint reward is 20% worse than that seed’s own best reward, relative to the magnitude of the best value. Values above 100% are visually capped at 100% so that severe collapses do not flatten the rest of the colour scale. If several checkpoints tie at the same best reward, only the **earliest** one is marked with a star, so the map uses the same checkpoint-selection rule as the checkpoint summary table.

How to read the map.

- Each **row** is one training seed.
- Each **column** is a scheduled checkpoint from 5k to 75k steps.
- A **star** marks the earliest scheduled checkpoint at which that seed attained its best scheduled-evaluation reward.
- Light cells remain close to the seed’s best performance.
- Darker orange/red cells are progressively further below that seed’s best performance.
- An early star followed by dark cells indicates deterioration after an early good policy.
- A star appearing late in training indicates that useful improvement continued later in the 75k budget.

Because every row is expressed relative to its own best checkpoint, this appendix is an **optimisation-stability diagnostic**, not a substitute for the absolute reward comparisons in Figures 3A and 3B.

Appendix Figure C1. How each seed moved relative to its own best scheduled checkpoint

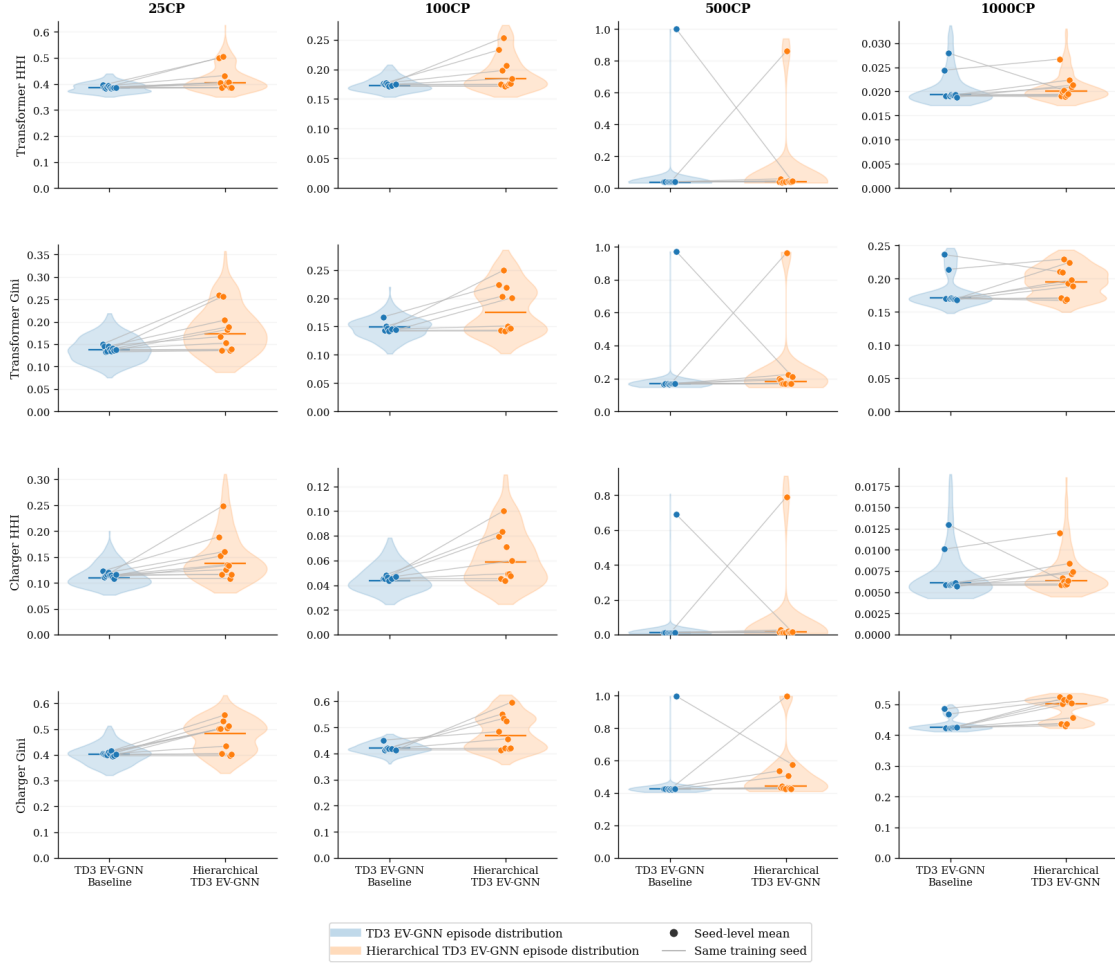


20 Appendix D - Allocation concentration distributions

The main text reports the mean HHI shift because it is the most direct concentration summary. This appendix retains the full transformer/charger **HHI** and **Gini** distributions so that the mechanism can be inspected without turning concentration into a performance score.

For each scenario and architecture, the translucent violin uses all 300 diagnostic episodes descriptively, while the overlaid points are the 10 seed-level policy means. Each scenario is given its own panel so that raw HHI/Gini values are compared **within scenario**, consistent with the entity-count caution in the main text.

Appendix Figure D1. Positive-charge allocation concentration within formal scenarios



20.1 Observed allocation-distribution pattern

The detailed distributions reinforce the HHI summary in the main text. At **25CP** and **100CP**, the hierarchical transformer and charger distributions are generally shifted toward greater concentration/inequality. At **500CP**, transformer HHI is lower under the hierarchy on average while charger-level concentration remains similar; isolated near-maximum outliers in both architectures show why the seed distribution matters. At **1000CP**, HHI distributions largely overlap, while charger Gini remains somewhat higher under the hierarchy.

The broad mechanism is therefore **not** “hierarchy always spreads allocation more evenly”. Its effect on allocation concentration depends on infrastructure level and formal scenario. HHI/Gini remain descriptive mechanism evidence and should not be interpreted as direct operational superiority.